

		Ministry of education Central STEM unit Capstone Calendar						
Week	Engineering	Students' work (teacher will)	what should be done in the Portfolio	what should be done in the Prototype	what should be done in the	Journal	Exams	Comments
1	•Present and Justify a Problem and Solution Requirements	•Introduce Egypt Grand Challenges, Capstone Challenge, Capstone Semester Dates; Introduce Capstone Portfolio template, Rubrics •Links for The shard files for the poster and portfolio •Teams Formed, Research problems and prior solutions •Students' representation skills •Document in the Capstone Portfolio (Egypt grand challenges, and problem to be solved) •The Poster design	• Each team is assigned a TEAM NUMBER •Each team's has an online shared Portfolio Name MUST follow this format Year-Semester Number-Team Number 2023-1-Team Number Egypt Grand Challenge(s) Which Egyptian Grand Challenge(s) are you addressing, and why are they important to address? Problem to be solved What specific problem are you addressing? What positive and negative consequences that are related to dealing with that problem?		Choose the design for the poster, the theme, the fonts, the title name, the logos and colors according to the rubrics	-	-	
2	• Present and Justify a Problem and Solution Requirements • Generating and Defending a Solution	•Research prior solutions, noting prior attempts strengths and weaknesses •Identify design requirements, Brainstorm possible solutions •Identify the design requirements that a good solution would meet •Document in the Capstone Portfolio (research, list of references -its place at the end of the file-, other solutions already tried) • Poster design •Students' representation skills	Which topics did you research about the problems? Students discuss factors that are related to the problem from different perspectives. Which topics did you research about the solutions? Students discuss technologies, scientific topics, or materials or any other factors that are related to the solution. Other Solutions Already Tried What are the solutions that have been applied trying to solve this problem? Why are the strengths and weakness of those solutions?	-	Improve the poster design and colors	-	-	
3	• Generating and Defending a Solution	• Select a solution and justify why it is a good solution. • Select which design requirements you will test in your prototype from among your other design requirements • Document in the Capstone Portfolio (Solution and Design Requirements, Selection of Solution) •the introduction and abstract in the poster • Students' representation skills	Solution and Design Requirements What characteristics (Solution Requirements) would a successful solution have, and how each of these characteristics can be met? Which design requirements did you choose and why did you chose it? Selection of Solution Describe the solution you decided to pursue and why did you choose it? Students will demonstrate the methods they will follow to solve the problem	design requirements give the stuents the chance to have an ieda about what the prototype should be tested for, what materials can be used. Which design requirements did you choose and why did you chose it?	Introduction Clearly and objectively identifies the problem and summarizes prior solution attempts strengths and weaknesses. Includes design requirements for a new solution that can be tested Summarizes how the team's solution was chosen and how it addresses design requirements Start editing the abstract Abstract is a brief description of the entire work described in the poster Abstract understandable without reading the entire poster Includes (1) purpose of the study, (2) brief statement of what was done (without including minor details of the methods), (3) brief statement of major findings, and (4) major conclusions		-	
4	• Generating and Defending a Solution	• Select a prototype for your team to design, build, and test the design requirements you identified for your prototype, • Document in the Capstone Portfolio (Selection of Prototype, Materials and Methods) • The poster (Materials and methods) • Create and submit a proposal for the Capstone Prototype • Capstone Budget Guidelines signed • Capstone Review Panel	Selection of Prototype Describe in details the prototype you have decided to construct, and how it will meet the design requirements you have chosen? In what ways can your prototype be tested? Materials and Methods List the materials you used to make your prototype. (Table: Item, quantity, description –If needed, usage, cost, source of purchase, picture) List the safety precautions you took to ensure your team is being safe.	The first design draft should be reviewed by the panel, give students the feedback, stuednts will make the needed modifications.	Materials and Methods Prototype materials lists and/or illustrations are summarized Describe how the prototype was constructed		-	
5	• Constructing and Testing a Prototype	• Follow Capstone Review Panel • Create a Test Plan to test whether your prototype meets your selected design requirements. • Review the safety precautions • Build your prototype. •Document in the Capstone Portfolio (Test plan) •The poster (review all written parts, continue improving the materials and methods)	Test Plan List the design requirements you have chosen to test with your prototype. List the steps for each test to conduct on the prototype (this test plan should be written in a way that supports repetition and testing by others)	make the needed modifications on the design. get the needed materials Start building the prototype.	Continue Materials and Methods A summary of Methods of the test plan for the prototype includes a summary of tests conducted and how they address design requirements		-	
6	• Constructing and Testing a Prototype	•Review the materials, methods and safety rules. •Build the prototype. •Rview all the written parts in the Capstone Portfolio •Rview all the written parts in the Capstone Poster	Improve the portfolio while working on building the prototype	Continue working on the prototype.	Improve the Poster while working on building the prototype		-	

7	• Constructing and Testing a Prototype	•Implement the test Plan. •Document the tests, the results, and any modifications in the Portfolio. •Revise the Prototype design based on test results (do as long as needed or until time runs out)	Data Collection List all data collected in your testing here What measurement tools did you use? What level of precision did you use in measurement? Provide a visual representation of your data and results. The best visual representations tell the story of your results without the need for additional writing.	implement the test plan (test the prototype according to the chosen design requirements)	Results All types of prototype testing results are presented, whether positive or negative Supporting documentation from the Capstone Portfolio is available to show data for tests or scenarios that were conducted for the prototype testing results Includes a table or figure that is appropriate for the type of prototype test results being described	-		
8		Review all the work and make sure that all team members have the same level of the understanding of what have been done>	Improve the portfolio while working on building the prototype	Continue working on improving the prototype.	Improve the Poster while working on building the prototype	-		
9	• Constructing and Testing a Prototype; Evaluation, Reflection, Recommendations	• Continue the implementation of the Test Plan and document • Continue to Revise Prototypes based on test results (do as long as needed or until time runs out). Prepare Capstone Poster for Exhibition using the rubrics and Capstone Poster Template. • Document in the Capstone Portfolio (Data collection) • The poster (improving the whole poster, continue writing the results) • Students' representation skills	Continue the Data Collection Improve the portfolio while working on building the prototype	Continue implementing the test plan after improving the prototype (test the prototype according to the chosen design requirements)	improve the Results part	-		
10	• EDP Progress: Evaluation, Reflection, Recommendations	• Analyze the Data—include analysis of the effectiveness of the design • Prepare your team's recommendation for future research, development written for others to replicate. • Document in the Capstone Portfolio (Analysis and discussion, recommendation, finalize the learning outcomes part) • The poster (Analysis, conclusion, recommendation) • Students' representation skills	Analysis and Discussion What is the analysis of the prototype behavior and the test plan results? What conclusions do you reach from the data you collected? (Your test and your measurements should be accurate enough to draw conclusions) (The data should demonstrate that the prototype met the identified design requirements) Recommendations What recommendations do you have for future work in this area? What would you tell another team who wanted to start where you stopped on your solution to help them? Learning Outcomes Which 10 discipline learning outcomes did you identify as related to your Design Challenge? Explain how each identified learning outcome was transferred to your Capstone.	Finalize the decorations and final look for the prototype if it needs	Analysis Discussion ties performance results (positive and negative) to the original challenge or problem being addressed and to the Grand Challenge Analysis ties the prototype testing results to the design requirements Analysis is supported by pictures, graphs, charts and other visuals, and prototype test results Analysis is supported by scientific laws and theories and provides evidence of learning transfer of discipline learning outcomes Conclusions Conclusions are drawn from the prototype test results and analysis. Recommendations Recommendations for future study are provided, including specific ways the project could be improved in the future	-		
11	• Finalize the Poster, the Portfolio, and the prototype.	• Finalizing and reviewing th Poster and the portfolio to be ready for submission. • Finalizing the Prototype for exhibition, (make sure thet all work is completed) • Experimental exhibition	review the research list at the end of the portfolio List the research sources you have used for your project in APA format. Review the whole portfolio.	Finalize the decorations and final look for the prototype if it needs	Literatures cited Includes only sources cited in the poster text (at least 5 credible sources) Different kinds of sources (paper- books, encyclopedia websites) not only websites includes only papers actually read by the students is prepared according to the American Psychological Association (APA) style guidelines Review the whole poster	-		
12	•Experimental exhibition						-	
13								
14								
15								
16						First term exams G10 & G11		
17								
اجازة نصف العام الدراسي								